

Explainability and Interpretability of LLMs in Finance

Lecture 12 — Explainable AI for Finance

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The Problem in One Scenario

THE BIGGER PICTURE

A small-business owner applies for a working-capital loan. A transformer-based credit system returns a *rejection* in seconds. Under the Equal Credit Opportunity Act, the loan officer must give the applicant **specific reasons**. The model offers only a probability — not a rationale.

Explainability = the degree to which a human can understand the cause of a model's decision (Miller, 2019).

Why finance is different

In most industries opacity is an inconvenience. In finance it is a **legal, fiduciary, and reputational** problem. Institutions must show that outputs are *appropriate, non-discriminatory, and auditable* — not merely accurate (SR 11-7).

LLMs sharpen the challenge: billions of parameters, uninterpretable activations, and outputs that depend on prompt phrasing in ways that are hard to characterise.

Legal Requirements: Four Frameworks

Framework	What it demands
EU AI Act (2024)	Credit scoring / insurance pricing = <i>high-risk</i> (Annex III). Transparency, input/output logging, human oversight; must explain what drove a <i>specific</i> decision.
SR 11-7 (Fed)	Model-risk management: document conceptual soundness; outputs explainable to non-technical risk managers; sensitivity & post-hoc attribution.
ECOA / Reg. B	Adverse-action notices must give the “principal reasons” (4–5 factors) in plain language. “Not creditworthy” is <i>not</i> enough.
GDPR Art. 22 / FCRA	Right to a “meaningful explanation” of the logic of solely-automated decisions.

Definition — Explanation vs. Interpretation

Interpretation: how does the model work *in general*? **Explanation:** why this output for *this* input? Regulation is primarily **explanatory** — specific decisions justified to specific persons (Doshi-Velez & Kim, 2017).

Three Audiences, Three Tolerances

Clients / applicants

- Actionable, plain-language, contrastive
- “DTI of 52% was 10 pp above threshold”
 $\succ$ “ x_{17} had SHAP -0.43 ”

Auditors / validators

- Both global + local
- Must **reproduce** explanations
- Non-deterministic XAI is unfit for audit

Regulators / examiners

- Higher technical rigour
- Global behaviour: disparate impact?
- Audit any case on complaint

Design tension

Methods that satisfy regulators may overwhelm retail clients; methods accessible to clients may be too simplified for auditors. A real strategy deploys **multiple explanation types in parallel**.

The Black-Box Problem for LLMs

Intrinsically interpretable classics:

- Logistic regression: score = weighted sum; weights *are* the interpretation.
- Decision tree: a readable set of binary-split rules.

LLMs are not interpretable in this sense:

- Knowledge in *distributed* representations across all layers — no parameter maps to a human concept.
- Attention $\neq$ causal explanation (later).
- Inputs can be *free text* $\Rightarrow$ effectively infinite input space $\Rightarrow$ exhaustive sensitivity analysis impossible.

Rudin (2019): stop explaining black boxes

For high-stakes decisions, prefer *intrinsically interpretable* models. **Rejoinder for finance:** a model that jointly reads financial ratios *and* qualitative business text may extract signal a numeric model cannot. Whether the extra signal is worth the interpretability cost is a *business/regulatory* judgement, not a technical one.

SHAP: Shapley Values from Cooperative Game Theory

THE BIGGER PICTURE

Treat each *feature* as a *player* in a cooperative game. A feature's attribution is its average marginal contribution across all orderings in which players join the coalition — a provably “fair” division of the prediction.

UNDER THE HOOD

For a game (N, v) with $v(\emptyset) = 0$, the Shapley value of player i is

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (n - |S| - 1)!}{n!} [v(S \cup \{i\}) - v(S)].$$

SHAP (Lundberg & Lee, 2017): players = features $x_1, \dots, x_n$; $v(S)$ = expected prediction with only S “known”. Then ϕ_i is how much feature i moves the prediction from the baseline $\mathbb{E}[f(X)]$.

SHAP: Three Properties Prized in Finance

The Shapley division is not just convenient — it is the *unique* attribution satisfying a set of axioms. Three of them matter directly for regulated credit work:

- **Efficiency:** $\sum_i \phi_i = f(x) - \mathbb{E}[f(X)]$ (accounts for the whole prediction).
- **Symmetry:** identical contributions $\Rightarrow$ equal values.
- **Dummy:** irrelevant features get zero.

Why this matters for ECOA

Efficiency guarantees the attributions *add up* to the decision: no part of the rejection is left unexplained. Symmetry and dummy give the fairness footing — features are treated consistently and irrelevant ones cannot be blamed.

SHAP for LLMs: A Credit-Decision Example

What is a “feature” in text? The **token**. But marginalising a token requires a reference: a mask token, a random vocabulary token, or sampling from a small reference LM.

TreeSHAP is exact in poly-time; for LLMs we *approximate* (KernelSHAP).

Fine-tuned BERT $\rightarrow P(\text{creditworthy}) = 0.14$ (baseline 0.50). KernelSHAP, 500 samples:

Token	SHAP
struggled	-0.089
declining	-0.073
down 30%	-0.061
competitive	-0.047
restaurant	-0.038
pandemic	-0.029
operates	+0.012

Negative tokens cover most of the $0.50 - 0.14 = 0.36$ gap.

$\Rightarrow$ partial ECOA statement: “primarily affected by language indicating declining revenue, operational struggle, and market difficulties.”

Caveat

SHAP measures *correlational* importance *within the model* — not real-world causation. A spurious correlation (jargon $\leftrightarrow$ protected class) is reported faithfully. Fairness auditing is a **separate** step.

LIME: Local, Model-Agnostic Surrogates

THE BIGGER PICTURE

A globally non-linear model is *locally* approximately linear. Fit a sparse interpretable model in the neighbourhood of the instance being explained.

Recipe: (1) perturb x by masking features $\rightarrow z_1, \dots, z_K$; (2) query $f(z_k)$; (3) fit sparse linear g weighted by proximity $\pi_x(z) = \exp(-D(x, z)^2 / \sigma^2)$.

UNDER THE HOOD

$$\xi(x) = \arg \min_{g \in \mathcal{G}} \underbrace{\sum_k \pi_x(z_k) (f(z_k) - g(z_k))^2}_{\mathcal{L}: \text{locality-weighted fidelity}} + \underbrace{\Omega(g)}_{\text{complexity}}.$$

Advantage: truly model-agnostic — only needs query access. Works on closed APIs (GPT-4, Claude) with *no* gradients.

Disadvantage: *instability* — stochastic perturbations $\Rightarrow$ different explanations on reruns. **Fix a seed** for audit reproducibility.

Attention Visualisation: Promise and Limits

Attention weight $\alpha_{ij} = \text{softmax}(\mathbf{QK}^\top / \sqrt{d_k})_{ij}$ normalises to one $\Rightarrow$ *tempting* to read as “token j drove the output.” Practitioners colour tokens by α .

Why this is weak

Jain & Wallace (2019): adversarial *alternative* attention distributions give *identical* predictions $\Rightarrow$ attention is not a unique explanation. **Wiegreffe & Pinter (2019):** attention can still be a useful *diagnostic*, not a causal account.

Definition — Faithfulness vs. Plausibility (Jacovi & Goldberg, 2020)

Faithful: accurately reflects the causal process producing the output. **Plausible:** looks convincing to humans, regardless of the true computation.

Attention is often *plausible* but *unfaithful* — dangerous: it can make an auditor certify a decision as well-reasoned when the real drivers differ. **Integrated Gradients** (Sundararajan, 2017) gives stronger faithfulness (needs gradients $\Rightarrow$ no closed APIs).

Chain-of-Thought as an Explanation Trace

THE BIGGER PICTURE

Instead of black-box post-hoc methods, exploit the LLM's own generation: have it emit a reasoning trace *before* the answer. The trace doubles as a plain-language explanation regulators can read and archive.

Dual function of CoT (Wei et al., 2022):

- 1 Improves decision quality on multi-step inference (e.g. covenant-breach checks).
- 2 The trace is a natural-language explanation for applicant + auditor.

UNDER THE HOOD

Definition — Explanation Trace. A sequence $r_1, \dots, r_T$ generated before output $\hat{y}$, each r_t following from prior steps and input x , together entailing $\hat{y}$. *Counterfactually faithful* if negating any r_t flips $\hat{y}$.

Chain-of-Thought: The Faithfulness Risk

A reasoning trace *looks* like an explanation. The danger is treating it as one without evidence that it reflects how the model actually decided.

Faithfulness risk

The model's real inference is parallel matrix computation — not sequential reasoning. A fluent trace may not reflect it. **Self-consistency** (Wang et al., 2022): sample K traces, take the majority answer $\Rightarrow$ more accurate, more consistent.

Practical stance for finance

Use CoT for its *decision-quality* gains and as a *human-readable* draft, but do not present the trace to a regulator as the causal account of the decision unless it has passed a counterfactual-faithfulness check.

Counterfactual Explanations: What Would Change It?

Different question: not “what drove the decision?” but “what is the closest state of affairs in which it flips?” **Actionable by construction** (Wachter et al., 2017).

“DTI of 54% was a key negative” $\prec$ “a ratio below 45% would likely lead to approval, all else equal.”

UNDER THE HOOD

$$x' = \arg \min_{z \in \mathcal{X}} d(x, z) \quad \text{s.t.} \quad f(z) = c, \quad c \neq f(x),$$

with d a cost-reflecting distance: weighted ℓ_1/ℓ_2 for structured inputs, edit / token-substitution distance for text.

Counterfactuals: Feasibility and Legality Constraints

Solving the optimisation is only half the job. The *closest* counterfactual is often not one a person can act on — or one a lender may legally cite.

Text makes this hard — and constraints matter

Discrete input space $\Rightarrow$ no direct gradient descent. Two-stage: identify sensitive dimensions, then articulate them (often *with* an LLM). The closest counterfactual may be **infeasible** (“if your age were different”) or **illegal** — age is protected under ECOA. Add feasibility & non-discrimination constraints.

The actionability payoff

Done right, a counterfactual turns a denial into guidance: “lower your DTI below 45%” is something an applicant can pursue, unlike a raw SHAP score. That is why it is the explanation form regulators and clients both tend to prefer.

Natural-Language Justifications: Separation of Concerns

Pipeline: (1) a primary model decides; (2) SHAP / IG identifies top drivers; (3) an LLM *translates* those drivers into plain-language prose for the target audience.

Why separate the three roles

Primary model	evaluated for <i>predictive accuracy</i>	Each component is independently testable and replaceable.
Attribution	evaluated for <i>faithfulness</i> to that model	
LLM generator	evaluated for <i>clarity, completeness, compliance</i>	

The generator must not hallucinate attributions

If SHAP said “revenue decline,” the generator must *not* add “poor personal credit history” — even if plausible. Feed attribution scores as **structured input**; constrain via prompt engineering or constrained decoding to refer *only* to provided attributions.

Case Study 1: Hybrid SME Credit-Scoring Pipeline

A fintech lender's three-stage pipeline:

- ➊ **Structured (42 vars):** gradient-boosted tree + *exact* TreeSHAP.
- ➋ **Unstructured text:** fine-tuned DistilBERT + KernelSHAP (1,000 samples) → top-5 negative tokens.
- ➌ **Adverse-action generation:** GPT-4o, constrained to the feed below.

You are a credit officer drafting a regulatory-compliant adverse action notice. The application was declined. The primary reasons, in order of importance, are:

```
1. Debt-to-income ratio: 54% (threshold 45%) - SHAP -0.18
2. Revenue trend: -28% YoY - SHAP -0.12
3. Description: "struggling","declining" - txt -0.09
4. Cash reserve: 0.8 mo (threshold 3) - SHAP -0.07
5. Industry: Hospitality (elevated default) - SHAP -0.05
Do NOT introduce factors not listed. Do NOT mention protected
characteristics. Limit to 150 words.
```

Principle: LLMs are safest in the explanation layer when inputs are *tightly constrained by upstream attribution*. The LLM transforms attributions into prose; it must not exercise credit judgement here.

Case Study 2: ECOA/FCRA-Compliant Denial Letters

Reg. B requires notice within **30 days**: (1) specific reasons, (2) right to a statement of reasons, (3) right to a free consumer-report copy if used. “Principal reasons” = typically 4–5, stated *specifically*.

Failure mode 1: vagueness

“credit profile did not meet criteria” — insufficient.
“DTI 54% exceeded our 45% max” — compliant.

Failure mode 2: protected factors

Demographic signals in text $\Rightarrow$ explanation may reference race, sex, age... Must filter.

Robust implementation:

- 1 Structured attribution feed — *non-protected* variables only.
- 2 System prompt forbidding new factors / protected characteristics.
- 3 Post-generation compliance check (classifier / keyword filter).
- 4 Human-review queue for any flagged letter.

Case Study 3: MiFID II Suitability Disclosures

MiFID II: investment firms must give retail clients a **suitability report** explaining why a product fits their objectives, risk tolerance, situation, and knowledge — *before* execution.

CFA Institute (2025) — three requirements for AI-generated disclosures:

- 1 Reasoning **traceable** to the client data on file.
- 2 Language must **not** overstate returns or understate risks.
- 3 **Auditable / reproducible**: same data + config → materially equivalent document.

Robo-adviser, $n = 12,000$ clients, annual review

Structured profile → risk tier + suitability score → Integrated Gradients → top-3 drivers → LLM prompted to use stated horizon & risk tolerance, acknowledge capital-loss risk, cite no return figure absent from the KID → compliance filter. **Quarterly 2% human sample**; illustrative revision rate $\approx 0.3\%$, all from objective-vs-product tension.

Faithfulness: Sufficiency and Comprehensiveness

The hard part: unlike predictions, there is *no ground-truth explanation* to benchmark against.

UNDER THE HOOD

Let $e(x)$ be the explained (important) features, $\bar{e}(x)$ its complement.

$$\text{Suff}(e, f, x) = 1 - |f(x) - f(x_{e(x)})| \quad (\text{keep only } e: \text{prediction preserved?})$$

$$\text{Comp}(e, f, x) = |f(x) - f(x_{\bar{e}(x)})| \quad (\text{remove } e: \text{prediction degraded?})$$

Suff $\rightarrow 1$ = perfect faithfulness; high Comp = the explained features really drove it.

How the methods score

SHAP	efficiency property $\Rightarrow$ a form of <i>sufficiency</i> for the full set
LIME	may <i>fail</i> sufficiency: deliberately sparse surrogate omits correlated features
Attention	typically <i>fails comprehensiveness</i> : ablating attended tokens often leaves outp

Completeness and Plausibility

Completeness

Does it account for *all* relevant drivers, not just the top one? An ECOA notice citing 1 of 5 significant reasons is incomplete.

For SHAP, directly measurable:

$$\frac{\sum_{i \in e(x)} \phi_i}{f(x) - \mathbb{E}[f(X)]}.$$

Plausibility

Would a domain expert agree it makes sense? “high DTI” = plausible; “the word *the*” = implausible (even if accurate).

Correlates with — but is *not* — faithfulness. Measured by agreement with expert-annotated ground truth, or readability for ECOA audiences.

Key distinction

A plausible explanation may be faithful *or* unfaithful. Plausibility alone is a poor quality gate.

Human-Subject Studies with Domain Experts

The ultimate regulatory test: can a qualified expert *use* the explanation to make appropriate decisions? Three standard designs (Doshi-Velez & Kim, 2017):

Forward simulation	Given an explanation, predict the model's output on a held-out input.
Decision support	Approve/reject with vs. without explanations — does it help catch model errors?
Trust calibration	Should <i>raise</i> trust on correct predictions, <i>lower</i> it on incorrect ones.

Practical constraint: credit officers / compliance specialists are expensive participants. Alternative: *vignette studies* with synthetic scenarios via professional networks.

The trap

Experts often find plausible-but-unfaithful explanations *just as satisfying* as faithful ones. Human evaluation alone is not a reliable gate — pair it with quantitative faithfulness metrics.

Five Takeaways

- 1 **Explainability is a legal requirement**, not an aspiration: EU AI Act, SR 11-7, ECOA, MiFID II.
- 2 **Classical XAI adapts to LLMs but not trivially**: SHAP & LIME need feature definitions + marginalisation/perturbation schemes; both can be unstable.
- 3 **Attention is not faithful** in general; gradient methods (Integrated Gradients) are stronger but need model internals.
- 4 **LLM-native methods** (CoT, counterfactuals, NL justifications) give plain language — but constrain the generator to upstream attributions and add compliance filters.
- 5 **Evaluate on multiple axes**: faithfulness, completeness, plausibility, human utility. No single metric suffices.

The recurring architecture

Decide → attribute (faithfully) → translate (constrained LLM) → filter (compliance) → human review. Auditable at every stage.

APPENDIX

Deeper Foundations, Benchmarks & Further Reading

Extra / optional material — beyond the core lecture.

Appendix A: Why Shapley Values Are “Fair”

The combinatorial weight $\frac{|S|! (n - |S| - 1)!}{n!}$ is the probability that feature i joins coalition S at this position in a uniform random ordering of all $n!$ permutations.

Axiomatic uniqueness (Shapley, 1953): the Shapley value is the *unique* attribution satisfying efficiency, symmetry, dummy, and additivity simultaneously.

Complexity:

- Exact Shapley: exponential in the number of features.
- **TreeSHAP** (Lundberg et al., 2020): *exact*, polynomial-time for tree ensembles.
- Neural nets / LLMs: KernelSHAP sampling approximation (its weighting connects directly to LIME’s locality kernel).

Appendix B: Marginalising a Token Is Hard

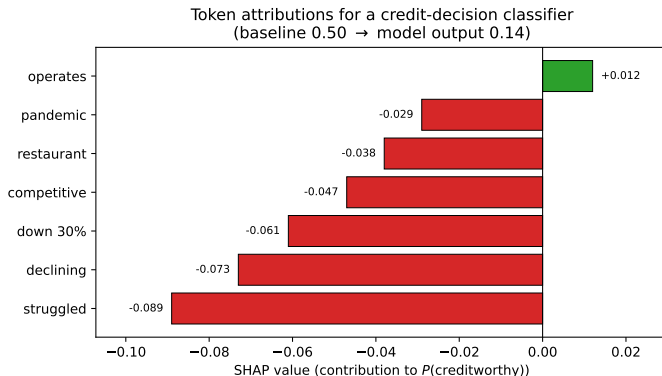
SHAP needs the model's prediction when a feature is "unknown." For tabular data, marginalise over the data distribution. For *discrete text*, the expectation over the token's marginal distribution given context requires a *language model itself*.

Three practical surrogates:

- ➊ Replace with a [MASK] token.
- ➋ Replace with a randomly sampled vocabulary token.
- ➌ Sample from a smaller reference language model.

Each choice changes the reference distribution — and hence the attribution. Document the choice for audit.

Appendix C: SHAP Attribution Figure (Credit Example)



Token-level SHAP attributions: red (negative) tokens push $P(\text{creditworthy})$ down from the 0.50 baseline toward 0.14; “operates” adds +0.012. Generated *deterministically* by `gen_shap_attribution.py` from the fixed example values — no live model or data.

Appendix D: Further Reading

- **Foundations:** Shapley (1953) axiomatic uniqueness; Lundberg & Lee (2017) unified additive attribution; Lundberg et al. (2020) TreeSHAP.
- **LIME:** Ribeiro et al. (2016) — readable, worked text/image/tabular examples.
- **Evaluation:** Doshi-Velez & Kim (2017) — vocabulary for human-subject XAI.
- **Counterfactuals:** Wachter et al. (2017) — includes GDPR Art. 22 context.
- **Human side:** Miller (2019) — what users actually want from explanations.
- **Attention debate:** Jain & Wallace (2019) *vs.* Wiegrefe & Pinter (2019); resolution by Jacovi & Goldberg (2020).
- **Regulation:** EU AI Act (2024); ESMA MiFID II suitability; Fed SR 11-7; CFA Institute (2025) industry survey; Arrieta et al. (2020) full taxonomy.

Within this course: Credit Risk (calibration, validation prerequisites for SHAP pipelines); LLM Agents (CoT as planning); Applications & Future (fairness / bias, tied to ECOA non-discrimination).